PUSHFORWARD OF A QUASI-COHERENT MODULE ALONG A CARTESIAN MORPHISM OF SIMPLICIAL SCHEMES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-describe-functoriality, lemma-check-cartesian-module, lemma-quasi-coherent-sheaf; external tools from Schemes and Cohomology of Schemes (flat base change).

Editorial note. The label lemma-pushforward-cartesian-module names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Componentwise description. Same unit, Lemma lemma-describe-functoriality. F_n = f_{n,*}G_n for the pushforward along a morphism of simplicial schemes.
- Cartesian criterion. Same unit, Lemma lemma-check-cartesian-module. F is cartesian iff the degeneracy base-change maps are isomorphisms.
- Quasi-coherence characterization. Same unit, Lemma lemma-quasi-coherent-sheaf. A module on the simplicial Zariski site is quasi-coherent iff componentwise quasi-coherent and cartesian.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Prerequisite pointer invoked at proof line(s) 6573; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 02N6: sources/stacks-chow/tags.txt, source lines 3469–3469. Prerequisite pointer invoked at proof line(s) 6566; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07TG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2474–2474. Prerequisite pointer invoked at proof line(s) 6577; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 09VP: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 414–414. Prerequisite pointer invoked at proof line(s) 6564; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D7M: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2817–2817. Prerequisite pointer invoked at proof line(s) 6570; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Prerequisite pointer invoked at proof line(s) 6568; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : V \to U$ be a cartesian morphism of simplicial schemes. Assume the morphisms $d^n_j : U_n \to U_{n-1}$ are flat and the morphisms $V_n \to U_n$ are quasi-compact and quasi-separated. For a quasi-coherent module $\mathcal{G}$ on V_{Zar} the pushforward $f_*\mathcal{G}$ is a quasi-coherent module on U_{Zar}.*

Proof. If $\mathcal{F} = f_*\mathcal{G}$, then $\mathcal{F}_n = f_{n,*}\mathcal{G}_n$ by Lemma 09VP. The maps $\mathcal{F}(\varphi)$ are defined using the base change maps, see Cohomology, Section 02N6. The sheaves $\mathcal{F}_n$ are quasi-coherent by Schemes, Lemma 01LC and the fact that $\mathcal{G}_n$ is quasi-coherent by Lemma 0D7M. The base change maps along the degeneracies d^n_j are isomorphisms by Cohomology of Schemes, Lemma 02KH and the fact that $\mathcal{G}$ is cartesian by Lemma 0D7M. Hence $\mathcal{F}$ is cartesian by Lemma 07TG. Thus $\mathcal{F}$ is quasi-coherent by Lemma 0D7M. □